

WITBANK, South Africa

WMO: 682730

Lat: 25.8338S

Lon: 29.1926E

Elev: 1555

StdP: 83.99

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.7	4.0	-12.9	1.5	15.0	-10.7	1.8	14.0	8.1	14.3	7.3	13.6	2.7	190	0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.9	30.8	16.3	29.5	16.2	28.3	16.1	19.8	26.3	19.2	25.1	18.7	24.2	3.7	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	15.7	21.6	17.4	15.1	21.1	17.0	14.7	20.8	64.2	26.4	62.0	25.2	60.1	24.3	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 8.0	6.9	6.1	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-2.2	20.5	1.1	0.8	-3.1	21.1	-3.7	21.6	-4.4	22.1	-5.2	22.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	20.8	20.8	19.6	17.2	15.0	12.5	12.1	14.6	18.2	19.0	19.5	20.6
	DBStd	4.02	2.12	1.97	2.32	2.27	2.64	2.39	2.28	3.20	3.44	3.06	2.81	2.46
	HDD10.0	22	0	0	0	0	3	7	7	4	1	0	0	0
	HDD18.3	760	4	3	13	46	106	176	192	119	44	29	20	8
	CDD10.0	2750	334	303	298	217	158	82	73	147	246	280	284	329
	CDD18.3	446	79	73	53	12	2	0	0	3	39	51	54	79
	CDH23.3	3082	473	391	276	76	41	3	2	59	372	476	425	490
	CDH26.7	679	116	80	42	7	3	0	0	1	86	129	98	118
Wind	WSAvg	2.8	3.0	3.0	2.8	2.5	2.3	2.5	2.5	2.7	3.1	3.3	3.2	2.9
Precipitation	PrecAvg	714	128	91	88	35	14	6	2	6	17	79	112	131
	PrecMax	1085	248	299	200	119	73	31	7	30	93	165	209	243
	PrecMin	452	57	10	23	0	0	0	0	0	0	30	31	62
	PrecStd	163	54	61	55	30	17	8	2	9	22	38	51	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.8	31.0	29.8	27.8	26.2	23.5	23.1	26.7	31.0	31.9	31.9	31.4
		MCWB	18.1	17.3	15.7	14.9	12.4	10.6	9.0	12.1	13.8	15.6	15.4	18.4
	2%	DB	29.6	29.0	27.9	25.5	24.2	21.5	21.3	25.1	29.1	29.8	29.4	29.5
		MCWB	17.9	17.4	15.8	14.4	11.8	9.7	9.0	11.4	13.3	14.6	15.7	17.8
	5%	DB	27.8	27.4	26.3	23.9	22.7	19.9	20.0	23.6	27.2	28.0	27.7	28.0
		MCWB	18.2	17.5	16.2	14.3	11.6	9.1	8.5	10.7	12.5	14.3	15.8	17.7
	10%	DB	26.0	25.9	24.9	22.4	21.0	18.3	18.4	21.8	25.5	26.0	26.0	26.2
		MCWB	17.8	17.3	16.3	14.0	11.3	8.8	8.5	10.0	11.9	14.2	16.0	17.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.9	20.6	19.5	17.8	14.2	12.2	11.2	13.9	15.8	17.9	19.2	20.1
		MCDB	27.6	27.3	24.4	23.4	22.1	20.4	18.6	24.5	25.9	27.1	26.1	27.8
	2%	WB	19.8	19.6	18.6	17.0	13.3	11.1	10.5	12.8	14.8	16.8	18.1	19.4
		MCDB	26.2	25.7	23.4	21.6	21.0	18.5	18.0	22.3	24.6	25.2	24.8	26.5
	5%	WB	19.2	18.9	17.9	16.1	12.7	10.4	9.8	11.9	14.0	16.1	17.5	18.7
		MCDB	24.5	24.5	22.5	20.3	20.2	17.3	17.5	20.7	23.9	23.8	24.2	25.2
	10%	WB	18.6	18.2	17.3	15.2	12.0	9.7	9.1	11.0	13.2	15.4	16.9	18.1
		MCDB	23.5	23.3	22.0	19.9	19.2	16.3	17.0	19.5	22.1	22.4	23.5	24.1
Mean Daily Temperature Range	5% DB	MDBR	9.9	10.1	10.0	10.6	12.3	12.7	13.0	13.7	13.5	12.7	11.3	10.1
		MCDBR	13.2	12.8	12.7	12.7	14.1	14.3	14.6	14.7	14.9	15.1	13.8	12.7
		MCWBR	4.6	4.0	4.3	4.8	5.7	6.3	6.4	5.9	5.4	5.1	5.0	4.4
	5% WB	MCDBR	10.2	10.7	9.6	10.4	12.7	12.5	13.1	13.7	14.0	12.7	11.7	10.8
		MCWBR	4.3	4.1	4.0	4.4	5.4	6.0	6.2	5.7	5.4	5.1	4.9	4.5
Clear-Sky Solar Irradiance	taub	0.390	0.390	0.380	0.335	0.298	0.279	0.281	0.336	0.380	0.416	0.378	0.385	
	taud	2.322	2.325	2.332	2.434	2.507	2.529	2.511	2.346	2.246	2.184	2.327	2.326	
	Ebn at Noon	954	938	916	915	912	911	922	899	901	902	958	960	
	Edh at Noon	138	134	127	104	88	82	86	112	136	153	136	138	
All-Sky Solar Radiation	RadAvg	6.60	6.31	5.56	4.77	4.38	3.98	4.31	5.03	5.92	6.32	6.59	6.70	
	RadStd	0.64	0.58	0.50	0.41	0.22	0.20	0.10	0.23	0.29	0.40	0.57	0.38	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	+0.66	+0.75	N/A	N/A	N/A	N/A	N/A	-181	+234	+87

Nomenclature: See separate page